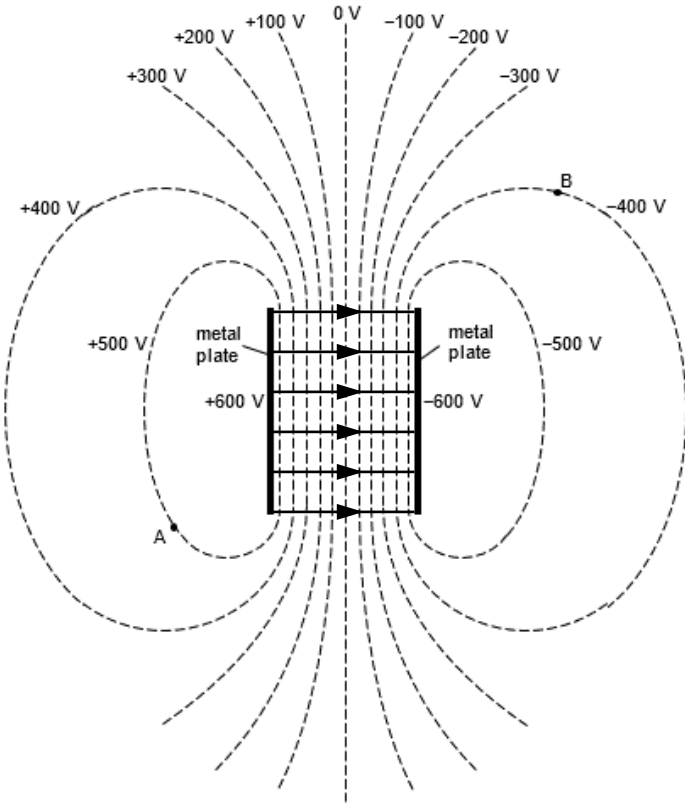


|                 |                                                                                                                             |            |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------|------------|
|                 | $= 2.0 \times 10^{-4} \text{ V m}^{-1}$                                                                                     | [1] answer |
| <b>3(a)(i)</b>  | The electric field strength $E$ at a point is the <u>negative of the potential gradient</u> $-\frac{dV}{dx}$ at that point. | [1]        |
| <b>3(a)(ii)</b> | The <u>equipotential lines are equally spaced</u> with the same potential difference between consecutive lines.             | [1]        |

| No.       | Solution                                                                                                                                                                                                                            | Remarks                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| 3(a)(iii) | <p>Since <math>E = \frac{V}{d}</math>, we have</p> $E = \frac{600 - (-600)}{2.5 \times 10^{-2}}$ $E = 4.8 \times 10^4 \text{ N C}^{-1}$                                                                                             | <p>[1] for correct substitution<br/>[1] for correct answer</p>                                              |
| 3(b)      |                                                                                                                                                  | <p>[1] for six uniform field lines between the plates<br/><br/>[1] for correct direction of field lines</p> |
| 3(c)(i)   | <p>Since the gain in electric potential energy of the particle is <math>q\Delta V</math>, we have</p> $q \times (-400 - 500) = 1.44 \times 10^{-16}$ $q = -1.60 \times 10^{-19} \text{ C}$                                          | <p>[1] for correct substitution<br/>[1] for correct answer</p>                                              |
| 3(c)(ii)  | The particle is an electron.                                                                                                                                                                                                        | <p>[1]<br/>No e.c.f. from (c)(i)</p>                                                                        |
| 3(c)(iii) | <p>By conservation of energy, loss in k.e. = gain in e.p.e.</p> $\frac{1}{2} \times 9.11 \times 10^{-31} \times \left[ (3.50 \times 10^7)^2 - v_f^2 \right] = 1.44 \times 10^{-16}$ $v_f \approx 3.01 \times 10^7 \text{ m s}^{-1}$ | <p>[1] for applying conservation of energy<br/>[1] for correct answer</p>                                   |

| No.     | Solution | Remarks     |
|---------|----------|-------------|
| 4(a)(i) | 400      | [1] for p.d |